

Biocatalysis Mechanism Composition: Catalysis Without Identities

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Abstract

A biochemical process is conventionally specified by naming its participants: which species, which complexes, which cofactors, in which order. We show that this specification contains a part that does no work. A process is fully determined by its *residue chain*—a finite sequence of steps in which the residue deposited by one step is the boundary condition of the next—together with, for each step, a single number in $[0, 1]$ recording the fraction of the remaining gap to the target that the step closes. Nothing else about a step enters. We call such a step a *rung*, such a sequence a *ladder*, and we prove that the identities are inert: two processes with the same power sequence are the same ladder, and no observable of the process distinguishes them. The account rests on two primitives, both derived here rather than assumed. First, individuation costs an irreducible floor $\beta > 0$, because drawing a boundary takes time during which the complement moves. Second, and consequently, there is an asymmetry between *probing* a region—entering it along a trajectory, which draws no boundary and is free—and *committing* it—individuating it, which costs at least β regardless of how improbable the region is. From these we obtain: composition is multiplicative, $\pi(\gamma_1 \cdots \gamma_n) = 1 - \prod_i (1 - \pi_i)$; repetition of a single rung has strictly diminishing returns and never reaches the target; a ladder closes its gap in the limit if and only if $\sum_i \pi_i$ diverges; and a structure that has been committed once is thereafter traversed by probes

alone, which is catalysis without consumption, obtained from the asymmetry and not from an energetic argument. We give a small-step operational semantics with progress, preservation and monotone commitment, and a static analysis that rejects at compile time a ladder whose declared target its declared rungs cannot reach. The framework is demonstrated on a chain of length four presented purely as a power sequence: we show that the conventional description of that chain is recoverable from no observable the formalism admits, and that every question usually asked of it—where the control lies, what a modification is worth, whether two chains are the same process—is answered from the powers alone. Classification and improvement of catalysts thereby become operations on numbers rather than on names.

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9	Static Analysis	11		Ask for a description of a biochemical pro- cess and you receive a list of participants in an order: these species, transformed by these agents, through these intermediates. The de- scription is detailed, it is checkable against experiment, and it is how the process is taught. It is also, we shall argue, partly inert—it con- tains material that no observable of the pro- cess depends on, and which can be deleted without loss.	
10	Cost	13		The claim is not that the identities are un- known, or unimportant for other purposes, or that a chemist should stop naming things. It is narrower and sharper: <i>given a process,</i> <i>the identities of its participants do not enter</i> <i>any question the process itself answers.</i> What enters is the sequence of changes and, for each change, how much of the remaining distance to the outcome it accomplishes. Two processes agreeing on that sequence agree on everything, however different their participants.	
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1.2 Why one would want this

Three practical questions become tractable when the identities are removed.

Classification. Asking whether two catalysts are “the same kind” currently means comparing mechanisms, folds, active-site residues, or sequence homology, and these criteria disagree with one another [Khersonsky and Tawfik, 2010]. On the present account two catalysts are the same rung if they have the same power, and there is nothing further to compare. The criterion is a single number and it is measurable.

Improvement. Asking how to improve a catalyst currently means proposing a modification and testing it. On the present account improvement is an increase in a rung’s power, and two results say where the headroom is. Theorem 5.3 rules out repetition: copies of what already works have geometrically diminishing returns. Theorem 10.3 then identifies the rung whose improvement buys the most, and its answer is not the one intuition supplies—see theorem 10.4.

Comparison across scales. A ladder is a ladder whether its rungs are molecular, cellular, or industrial. The composition law (theorem 5.1) does not know what a rung is made of, so processes that share no chemistry can be compared directly.

1.3 What has to be derived

We take nothing from a prior formalism. Two things must be established before the account can be stated:

- (i) **The floor.** That individuating anything costs an irreducible positive amount. This is section 2, and it is derived from the observation that drawing a boundary takes time during which what lies outside it changes.
- (ii) **The asymmetry.** That entering a region and individuating a region are operations of different kinds, the first free

and the second floored. This is section 3. It is what makes reuse possible, and hence what makes catalysis possible.

Everything else—rungs, ladders, composition, the semantics, the static analysis—follows.

1.4 Organisation

Part II derives the floor and the asymmetry. Part III defines rungs and ladders and proves the composition results. Part IV contains the central theorem: the identities do not enter. Part V gives the operational semantics, the static analysis, and the cost model. Part VI works a chain of length four entirely in powers. Part VII states the predictions and the computational protocol.

Part II Substrate

2 The Floor

2.1 Individuation takes time

Postulate 1 (Existence by distinction). An item exists as a distinct entity if and only if there is a well-defined complement from which it is distinguished. Its existence consists in the distinction and not in any property it possesses independently of it.

Postulate 2 (Distinction takes time). Establishing a distinction requires a finite non-zero interval $\tau > 0$, during which the complement evolves.

Remark 2.1. Postulate 2 is forced, not chosen. Were $\tau = 0$, the state of the world immediately before the distinction and immediately after would coincide; but a distinction that changes nothing has not been drawn. Hence $\tau > 0$, and the second clause follows for any world that is not static.

Definition 2.2 (Boundary thickness). Let A be individuated over $[t_0, t_0 + \tau]$. Its *boundary thickness* is the measure of the set of states which, at completion, belong determinately neither to A nor to its complement.

Theorem 2.3 (Thickness is positive). *Any individuation carried out under Postulates 1 and 2 has strictly positive boundary thickness.*

Proof. By Postulate 2 the operation occupies $\tau > 0$ and the complement evolves during it. Consider a state lying outside A at t_0 and inside the region assigned to A at $t_0 + \tau$, or conversely; such states exist whenever the complement is non-static, which Postulate 2 guarantees. For such a state the assignment appropriate at the start and the assignment appropriate at completion disagree, so it is determinately in neither part. The set of such states is non-empty and its measure is the thickness. $\square$

2.2 The floor is uniform when refinement is capped

Theorem 2.3 gives positivity separately for each individuation. The results below need a bound holding uniformly, and positivity of each member of a family does not give positivity of the infimum: the numbers $1/n$ are all positive with infimum zero. A hypothesis is therefore required, and we state it rather than let it pass unnoticed.

Definition 2.4 (Contact graph). A *contact graph* is $\mathcal{G} = (V, E, w)$ with V finite and non-empty, E a set of unordered pairs of distinct vertices, and $w : E \rightarrow \mathbb{R}_{>0}$ [Diestel, 2010]. A distinguished vertex $\mathbf{m} \in V$, the *medium*, is adjacent to every other vertex; the remainder are *items*. The weight of a contact is the cost of maintaining the distinction between its endpoints.

Definition 2.5 (Cut, residue, floor). For $S \subseteq V$ containing at least one item and omitting $\mathbf{m}$, the *cut* is $\text{cut}(S) = \{\{u, v\} \in E : u \in S, v \notin S\}$ and the *residue* is $\text{res}(S) = \sum_{e \in \text{cut}(S)} w(e)$. The *floor* is $\beta = \min_{e \in E} w(e)$ and the *total* is $\Omega = \sum_{e \in E} w(e)$.

Lemma 2.6 (Positive floor, positive residue). $\beta > 0$, and every admissible cut satisfies $\text{res}(S) \geq \beta > 0$.

Proof. E is finite and non-empty and every weight is strictly positive, so β is the minimum of a finite non-empty set of positive reals and is

positive. Since S contains an item v and omits $\mathbf{m}$, and $\{v, \mathbf{m}\} \in E$ because $\mathbf{m}$ is adjacent to every item, $\text{cut}(S)$ is non-empty. Its weight is a sum of terms each at least β . $\square$

Remark 2.7 (Where finiteness does the work). Theorem 2.6 is a statement about a *finite* graph, and the finiteness is not decoration. A system able to refine without limit generates thicknesses with infimum zero, and for such a system no uniform floor exists. Every result downstream inherits this hypothesis. We verify both branches numerically in section 14, because a hypothesis that is never tested against its own failure case is not a hypothesis but a habit.

Definition 2.8 (Alignment). Fix a target region x^* . The *alignment* of a region x to x^* is the minimum residue separating them,

$$\alpha(x, x^*) = \min_{S \ni x, x^* \notin S} \text{res}(S), \quad (1)$$

computable in polynomial time by maximum flow [Ford and Fulkerson, 1956, Stoer and Wagner, 1997]. The *normalised alignment* is $a(x, x^*) = \alpha(x, x^*)/\Omega$.

Alignment measures distance-to-target as separation cost. A region far from the target is expensive to separate from it; a region at the target is separated only at the floor.

Lemma 2.9 (Alignment is floor-bounded). $a(x, x^*) \geq \beta/\Omega > 0$ for every x .

Proof. Immediate from theorem 2.6 applied to the minimising cut. $\square$

3 The Asymmetry

We now derive the distinction on which everything else rests.

Definition 3.1 (Probe and commit). Let T be a measure-preserving traversal of the state space [Walters, 1982]. A region W is *probed* when some iterate of T enters it: this is a fact about a trajectory and involves no boundary. A region W is *committed* when a boundary is drawn individuating W from its complement: this is an act, and by theorem 2.3 it deposits residue.

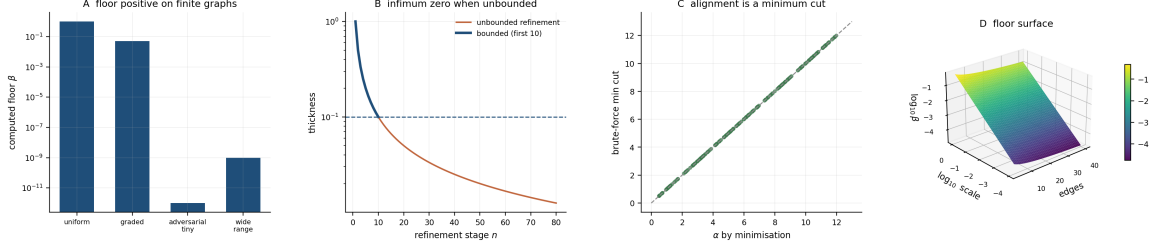


Figure 1: **The floor, the hypothesis it requires, and alignment as a minimum cut.** (A) Floor β computed—not assumed—as the minimum edge weight over four finite contact graphs spanning twelve orders of magnitude, including an adversarial instance whose smallest weight is 10^{-12} . Every instance returns a strictly positive floor. Positivity is a property of finiteness and strict positivity of the weights, not of their scale. (B) The failure branch. A system permitted to refine without limit generates thicknesses $1/n$ (orange): each is strictly positive and the infimum is zero, with every ε in a ladder down to 10^{-9} eventually cleared. Truncating the same sequence to a finite system (blue, first ten stages) restores a positive floor at 0.1 (dashed). The uniform floor therefore requires finiteness, and this chart is included so that the requirement is visible rather than tacit. (C) Alignment computed as a minimum residue over admissible cuts, against an independent brute-force enumeration of every bipartition, on forty randomly generated graphs. Points lie on the identity to machine precision. This is a check of the implementation against ground truth, not a restatement of the definition. (D) Floor surface over edge count and weight scale. The surface is positive everywhere on the domain and descends toward zero only as the scale does; increasing the edge count at fixed scale lowers the floor but never abolishes it, which is the content of the minimum over a finite set.

The two relations are of different kinds. Probing relates a trajectory to a point; committing relates an agent to a region. At every instant a trajectory occupies exactly one position, probed because the traversal took it there, with nothing committing anything.

Theorem 3.2 (Probe–commit asymmetry). *Let W be a region of positive measure $p(W) > 0$ in a finite-measure space, and T a measure-preserving traversal. Then*

- (i) **Committing costs the floor.** *Any commitment of W deposits residue at least $\beta > 0$, and this bound does not depend on $p(W)$.*
- (ii) **Probing is free.** *Almost every trajectory enters W , and does so without any boundary being drawn and without residue.*
- (iii) **Rarity affects frequency only.** *The long-run frequency with which W is probed equals $p(W)$. A rare region*

is probed as freely as a common one, merely less often.

Proof. (i) A commitment individuates W , so by theorem 2.3 its boundary has positive thickness, and by theorem 2.6 the associated cut has residue at least β . Neither step refers to $p(W)$.

(ii) By the Poincaré recurrence theorem [Poincare, 1890, Walters, 1982], for a measure-preserving T on a finite-measure space and any W with $p(W) > 0$, almost every point of W returns to W infinitely often, and by ergodicity of the induced map almost every point of the space enters W . Entry is a property of the trajectory. Theorem 3.1 attaches no boundary to it, and by theorem 2.3 residue arises only from boundaries.

(iii) The Birkhoff ergodic theorem gives, for almost every initial point, $\lim_n \frac{1}{n} \sum_{k < n} \mathbf{1}_W(T^k x) = p(W)$. This is a statement about frequency and carries no cost term. $\square$

Corollary 3.3 (Reuse is possible). *A structure that has been committed once thereafter renders the region it individuates available to probing, and every subsequent probe of that region is free. Such a structure is not consumed by the probes that traverse it, and may be traversed without limit.*

Proof. The commitment is performed once and deposits residue once, by theorem 3.2(i). Thereafter the region is individuated and a trajectory entering it is probing, by theorem 3.1. By theorem 3.2(ii) probing deposits nothing. Since nothing is deposited, nothing is spent, and the number of traversals is unbounded. $\square$

Remark 3.4 (This is catalysis, and it is not an energetic argument). Theorem 3.3 is the formal content of the observation that a catalyst is not consumed. It does not appeal to free energy, to a cycle that regenerates the catalyst, or to a convention that catalysts are excluded from stoichiometry. The catalyst is never spent because what passes through it probes rather than commits, and probing is free by theorem 3.2(ii). The asymmetry is doing all the work, and it was derived from the floor, which was derived from the fact that individuation takes time.

Part III Ladders

4 Rungs

Definition 4.1 (Rung and power). A *rung* toward target x^* is a partial map γ on regions whose application commits a contact and yields $\gamma(x)$. Its *power* at x is the fraction of the above-floor alignment that it closes,

$$\pi(\gamma, x) = \frac{a(x, x^*) - a(\gamma(x), x^*)}{a(x, x^*) - \beta/\Omega} \in [0, 1]. \quad (2)$$

A rung is *inert* if $\pi = 0$ and *absolute* if $\pi = 1$.

Lemma 4.2 (Power is bounded). $\pi(\gamma, x) \in [0, 1]$ for every rung, region and target.

Proof. By theorem 2.9, $a(\gamma(x), x^*) \geq \beta/\Omega$, so the numerator of eq. (2) never exceeds the denominator and $\pi \leq 1$. A map that increases alignment is not a rung toward x^* ; by convention such a map is assigned $\pi = 0$, giving $\pi \geq 0$. $\square$

Remark 4.3 (What a rung is not). Theorem 4.1 mentions no property of γ beyond the alignment it closes. It does not say what γ is made of, how it operates, how long it takes, or what passes through it. Two maps with the same power at x are, at this resolution, the same rung. That this is not a loss of information is the content of theorem 7.1.

Definition 4.4 (Ladder). A *ladder* is a finite sequence of rungs $\gamma_1, \dots, \gamma_n$ applied in order, each to the output of the last. Its *power sequence* is $(\pi_1, \dots, \pi_n)$ with $\pi_i = \pi(\gamma_i, x_{i-1})$ and $x_i = \gamma_i(x_{i-1})$.

5 Composition

Theorem 5.1 (Multiplicative composition). *A ladder with power sequence $(\pi_1, \dots, \pi_n)$ has composite power*

$$\pi(\gamma_1 \cdots \gamma_n) = 1 - \prod_{i=1}^n (1 - \pi_i). \quad (3)$$

Proof. Write $r_i = a(x_i, x^*) - \beta/\Omega$ for the above-floor gap after i rungs. By theorem 4.1, $\pi_i = (r_{i-1} - r_i)/r_{i-1}$, hence $r_i = (1 - \pi_i)r_{i-1}$. By induction $r_n = r_0 \prod_{i=1}^n (1 - \pi_i)$. The fraction of the initial gap closed by the ladder is $1 - r_n/r_0$, which is eq. (3); this is the power of the ladder regarded as a single rung, per theorem 4.1. $\square$

Corollary 5.2 (Diminishing returns). *Repeating one rung of power $\pi \in (0, 1)$ n times gives composite power $1 - (1 - \pi)^n$: strictly increasing in n , tending to 1, and never equal to 1 for finite n . The absolute share of the original gap closed by the n -th repetition is $\pi(1 - \pi)^{n-1}$, strictly decreasing in n .*

Proof. Immediate from eq. (3) with all $\pi_i = \pi$. For the second claim, the share closed by repetition n is $r_{n-1} - r_n = \pi r_{n-1} = \pi(1 - \pi)^{n-1}r_0$, and $(1 - \pi)^{n-1}$ is strictly decreasing since $0 < 1 - \pi < 1$. $\square$

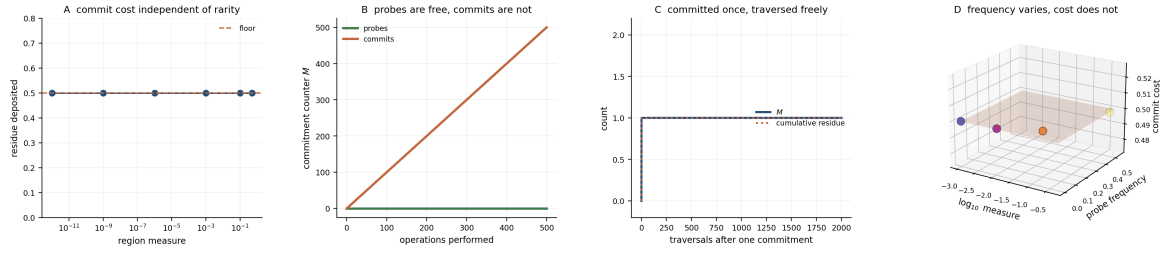


Figure 2: **Probing is free, committing is floored, and a structure committed once is thereafter traversed at no cost.** (A) Residue deposited by a commitment against the measure of the region committed, over six decades of measure. The deposit is constant and sits at the floor (dashed): the cost of individuating a region does not depend on how improbable that region is. This is the first clause of the asymmetry and it is the clause that makes rarity irrelevant to cost. (B) The commitment counter M under the two operations. Commits increment it one for one; probes leave it flat at zero however many are performed. The counter is the logical clock of the semantics, and the two lines are the operational content of the probe/commit distinction. (C) Reuse. A single commitment is performed, after which two thousand traversals are executed. M rises once and then stays; cumulative residue rises once and then stays. The structure is not consumed by the traversals, and no energetic argument enters the computation—only the asymmetry. (D) Probe frequency against region measure, with commitment cost as the vertical coordinate and the constant-cost plane overlaid. Frequency tracks measure over four decades while cost does not vary at all: rarity affects how often a region is visited and not what visiting it costs.

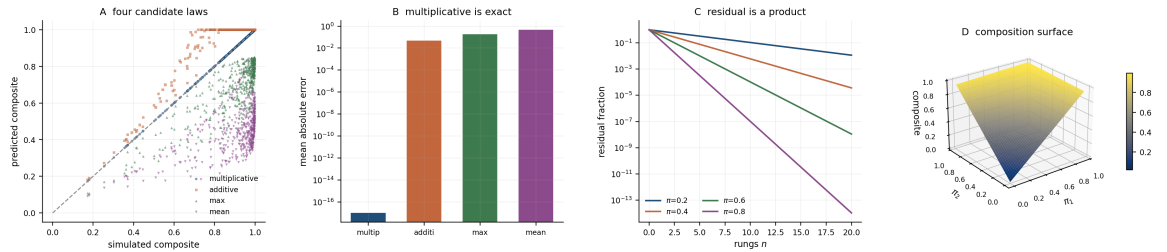


Figure 3: **Composition is multiplicative, and the alternatives are excluded by comparison rather than by assertion.** (A) Predicted against simulated composite power for four candidate composition laws on five hundred randomly generated ladders. The multiplicative law lies on the identity; the additive law saturates and overshoots, the maximum law systematically underestimates, and the mean law fails badly. The comparison is the test: scoring only the favoured law would not have distinguished it from the others. (B) Mean absolute error by law over four thousand ladders, on a logarithmic axis. The multiplicative law is exact to machine precision; the alternatives err by 0.047, 0.179 and 0.463 respectively. The gap spans sixteen orders of magnitude and is not a matter of fit quality. (C) Residual fraction against ladder length for four rung powers, on a logarithmic axis. Each trace is a straight line, which is the graphical content of the residual being a product $\prod_i (1 - \pi_i)$ rather than a sum. (D) Composite power over two rung powers. The surface is $1 - (1 - \pi_1)(1 - \pi_2)$: it rises to unity along either edge and is symmetric under exchange, so order does not matter, and it is strictly below the additive prediction everywhere in the interior.

Corollary 5.3 (Diversify, do not repeat). *Let a target power $\pi^* < 1$ be required. Using only copies of a rung of power π requires*

$$n \geq \frac{\log(1 - \pi^*)}{\log(1 - \pi)} \quad (4)$$

copies, and the requirement is unsatisfiable for $\pi^ = 1$ at any finite n . Reaching a demanding target economically requires rungs of differing power.*

Proof. From theorem 5.2, $1 - (1 - \pi)^n \geq \pi^*$ iff $(1 - \pi)^n \leq 1 - \pi^*$, and taking logarithms (both sides positive, $\log(1 - \pi) < 0$) reverses the inequality to give eq. (4). For $\pi^* = 1$ the right side is infinite. $\square$

Theorem 5.4 (Saturation dichotomy). *A ladder with powers $(\pi_i)_{i \geq 1}$, each in $[0, 1]$, drives the residual gap to zero in the limit if and only if $\sum_{i \geq 1} \pi_i = \infty$.*

Proof. The residual fraction after n rungs is $\prod_{i \leq n} (1 - \pi_i)$, which tends to zero iff $\sum_{i \leq n} -\log(1 - \pi_i) \rightarrow \infty$. Since $-\log(1 - \pi) \geq \pi$ for $\pi \in [0, 1]$, divergence of $\sum_i \pi_i$ implies divergence of $\sum_i -\log(1 - \pi_i)$. Conversely, if $\sum_i \pi_i < \infty$ then $\pi_i \rightarrow 0$, and $-\log(1 - \pi) \sim \pi$ as $\pi \rightarrow 0$, so the two series converge together by comparison. The dichotomy has the form of the Borel–Cantelli lemmas [Borel, 1909, Cantelli, 1917, Williams, 1991]. $\square$

Remark 5.5 (The dichotomy is a statement about the tail). The hypothesis $\pi_i \in [0, 1]$ excludes absolute rungs, and the exclusion is not a technicality. A single rung of power 1 closes the entire above-floor gap at one application, so any ladder containing one reaches zero residual gap whatever follows, for a reason having nothing to do with the convergence of $\sum_i \pi_i$. The validation of section 14 therefore indexes its test series from $i = 2$, so that the dichotomy is exhibited by tail behaviour rather than by a degenerate first step.

Part IV The Identities Are Inert

6 Processes

Definition 6.1 (Propagation). A *propagation* of x toward x^* is a finite sequence of contact commitments $S_1, \dots, S_n$ in which the residue deposited by S_i is the boundary condition of S_{i+1} . The propagation is *admissible* if it converges on x^* ; its intermediate states are otherwise unconstrained.

Theorem 6.2 (Intermediates are unconstrained). *An intermediate state of an admissible propagation may be arbitrarily far from the target without rendering the propagation inadmissible. Inadmissibility arises exclusively from failure to converge.*

Proof. Admissibility is defined in theorem 6.1 as a property of the terminal state alone. Let $S_1, \dots, S_n$ converge on x^* and let x_k be any intermediate. Replacing x_k by any x'_k from which the remaining rungs still converge yields a propagation satisfying the same terminal condition, hence equally admissible. No condition on $a(x_k, x^*)$ is imposed, so the value may be arbitrarily large subject only to convergence of the remainder. $\square$

Definition 6.3 (Process). A *process* is a propagation together with its power sequence. We write $\mathcal{P} = \langle (\pi_1, \dots, \pi_n), x_0, x^* \rangle$.

6.1 Observables

To state that identities do not enter, we must say what may be observed. We admit exactly the quantities the substrate defines.

Definition 6.4 (Admissible observable). An *admissible observable* of a process is a function of the alignments $a(x_i, x^*)$ for $0 \leq i \leq n$, the floor β , the total Ω , and the commitment count n .

Remark 6.5 (Why this is the right class, and what it excludes). Theorem 6.4 admits every quantity the substrate of part II can express: separation costs, the floor, the number

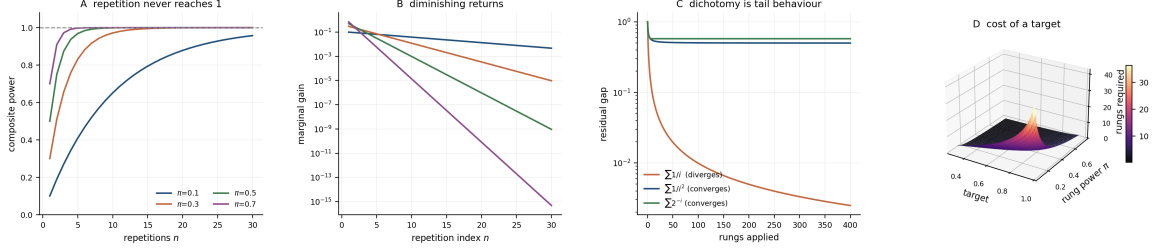


Figure 4: **Repetition has diminishing returns; the dichotomy between reaching and plateauing is a property of the tail.** (A) Composite power under repetition of a single rung, for four rung powers. Each curve rises strictly, tends to unity (dashed), and reaches it at no finite n . A ladder of near-duplicate rungs therefore cannot attain a demanding target however long it is made. (B) Marginal contribution of the n -th repetition, $\pi(1 - \pi)^{n-1}$, on a logarithmic axis. The decline is geometric. The stronger the rung, the larger the first contribution and the faster the decay—so repetition is least useful precisely where each copy is individually most effective. (C) Residual gap against rungs applied for three power series, indexed from $i = 2$ so that no rung is absolute and the dichotomy is exhibited by tail behaviour rather than by a degenerate first step. The divergent series $\sum 1/i$ drives the gap steadily toward zero; the convergent series $\sum 1/i^2$ and $\sum 2^{-i}$ plateau at finite residual and do not improve thereafter. (D) Rungs required to reach a target, over target and rung power. The surface diverges as the target approaches unity and as the rung power approaches zero, and is low only in the corner where strong rungs meet modest targets. This is the cost model: what a target costs is set jointly by how demanding it is and by how much each step accomplishes.

of commitments, and anything computed from them—composite power, residual gap, rung count, the position of the rate-limiting step. It excludes labels attached to rungs from outside. The theorem below says that this exclusion costs nothing, because no admitted observable can detect the labels. It does not say that labels are meaningless, only that a process does not depend on them.

7 The Inertness Theorem

Theorem 7.1 (Identities are inert). *Let $\mathcal{P}$ and $\mathcal{P}'$ be processes with the same power sequence, the same initial alignment, and the same floor and total. Then every admissible observable takes the same value on both. In particular no admissible observable separates them, whatever their participants.*

Proof. By theorem 5.1 the above-floor gap after i rungs is $r_i = r_0 \prod_{j \leq i} (1 - \pi_j)$, determined by r_0 and $(\pi_1, \dots, \pi_i)$ alone. The alignment is recovered as $a(x_i, x^*) = r_i + \beta/\Omega$, so the entire alignment sequence is determined by

the power sequence, the initial alignment, the floor and the total. The commitment count is the length of the power sequence. By theorem 6.4 an admissible observable is a function of exactly these data. Two processes agreeing on them therefore agree on every admissible observable. $\square$

Corollary 7.2 (Same power, same rung). *Two rungs with equal power at a given region are indistinguishable by any admissible observable, and may be substituted for one another in any ladder without altering any admissible observable of that ladder.*

Proof. A rung enters theorem 7.1 only through its power. Substituting one for another of equal power leaves the power sequence unchanged, hence by theorem 7.1 leaves every admissible observable unchanged. $\square$

Corollary 7.3 (Classification without identity). *The equivalence classes of rungs under indistinguishability are indexed by $\pi \in [0, 1]$. Classification of catalysts requires the measure-*

ment of one real number per catalyst and no structural information.

Remark 7.4 (What the theorem does and does not license). Theorem 7.1 is a statement about a fixed level of resolution. It says that at the resolution the substrate defines, identities are not detectable. It does not say that no finer resolution exists, nor that identities are causally idle in the world; a rung’s power is presumably a consequence of what the rung is. The claim is that the consequence is all that enters, so a description carrying the cause as well carries something the process does not use. Refining the resolution—adding observables—would change the class in theorem 6.4 and could break the theorem; we therefore state the observable class explicitly rather than leaving it implicit, so that the boundary of the claim is visible.

Proposition 7.5 (The converse fails). *Equality of every admissible observable does not entail equality of power sequences. Ladders with different power sequences may share a composite power.*

Proof. Take $(\pi_1, \pi_2) = (0.5, 0.5)$ and $(\pi'_1, \pi'_2) = (0.75, 0)$. Both give composite power 0.75 by eq. (3). Their intermediate alignments differ, so the full observable sequence distinguishes them; but an observer recording only the composite power cannot. The proposition therefore concerns the coarser observable and marks the limit of what composite power alone determines. $\square$

Remark 7.6. Theorem 7.5 is included because it is the natural objection to theorem 7.1 and it is correct as far as it goes. The theorem says the power sequence determines the observables; it does not say a single summary statistic does. An experiment measuring only end-to-end efficiency does not determine the ladder, and section 11 shows what further measurement is required.

Part V

The Machine

8 Operational Semantics

We give a small-step semantics in the style of Plotkin [1981], with soundness in the syntactic form of Wright and Felleisen [1994].

Definition 8.1 (Configuration). A configuration is $\mathcal{K} = \langle \mathcal{G}, M, x, e \rangle$: the contact graph, the commitment counter $M \in \mathbb{N}$, the current region, and the expression under evaluation.

Definition 8.2 (Step). A *step* is a commitment. It (a) commits a boundary in $\mathcal{G}$, (b) increments M by exactly one, and (c) deposits residue at least β . The counter is a logical clock [Lamport, 1978]. A probe is not a step: it changes the region but leaves $\mathcal{G}$ and M unchanged.

The reduction rules are:

$$\frac{}{\langle \mathcal{G}, M, x, \text{probe} \rangle \rightarrow \langle \mathcal{G}, M, T(x), \text{done} \rangle} \text{ (E-PROBE)}$$

$$\frac{\text{res}(S) \geq \beta \quad S \text{ admissible}}{\langle \mathcal{G}, M, x, \text{commit}(\gamma) \rangle \rightarrow \langle \mathcal{G} \oplus \text{cut}(S), M+1, \gamma(x), \text{done} \rangle} \text{ (E-COMMIT)}$$

$$\frac{a(x, x^*) - \beta/\Omega > \varepsilon}{\langle \mathcal{G}, M, x, \text{climb}(\gamma_1 \cdots \gamma_n) \rangle \rightarrow \langle \mathcal{G}, M, x, \text{commit}(\gamma_1); \text{climb}(\gamma_2) \rangle} \text{ (E-CLIMB)}$$

$$\frac{a(x, x^*) - \beta/\Omega \leq \varepsilon \quad \text{or} \quad n = 0}{\langle \mathcal{G}, M, x, \text{climb}(\cdots) \rangle \rightarrow \langle \mathcal{G}, M, x, \text{result} \rangle} \text{ (E-HALT)}$$

where **result** is Reached if the gap is within ε and Short(r_n) otherwise.

Definition 8.3 (Typing). Types are $\tau ::= \text{Region} \mid \text{Rung} \mid \text{Ladder} \mid \text{Result}$. The judgement $\Delta \vdash e : \tau @ \beta / \pi$ reads: e has type τ under ambient floor $\beta > 0$ and carries declared power π .

Theorem 8.4 (Progress). *If $\Delta \vdash e : \tau @ \beta / \pi$ and e is not a value, some reduction applies.*

Proof. By cases on e . For **probe**, (E-PROBE) applies unconditionally. For **commit**, the side condition $\text{res}(S) \geq \beta$ holds by theorem 2.6 whenever S is admissible, and admissibility is guaranteed by the typing of γ , so (E-COMMIT) applies. For **climb**, the guard of (E-CLIMB) and the guard of (E-HALT) are complementary and exhaustive, so exactly one applies. $\square$

Theorem 8.5 (Preservation). *If $\Delta \vdash e : \tau @ \beta / \pi$ and $K \rightarrow K'$, then $\Delta \vdash e' : \tau @ \beta / \pi'$ for some π' , with β unchanged and $M' \geq M$.*

Proof. Each rule preserves the type of its subject: (E-PROBE) maps a region to a region; (E-COMMIT) yields the region (RUNG) assigns; (E-CLIMB) leaves a **climb** in place; (E-HALT) produces **Result**. The floor is a component of Δ and no rule modifies it. M is incremented by (E-COMMIT) and left alone by the others, so $M' \geq M$. Reduction mutates $\mathcal{G}$; preservation holds against the mutated graph because $\mathcal{G} \oplus \text{cut}(S)$ retains the medium and all positive weights, so theorem 2.6 continues to apply. $\square$

Corollary 8.6 (Soundness). *A well-typed program does not get stuck, and every commitment it performs deposits residue at least $\beta > 0$.*

Theorem 8.7 (Monotone commitment). *M never decreases. Re-evaluating a syntactically identical commitment is a new commitment at strictly higher M , never a cached retrieval.*

Proof. No rule decrements M , and (E-COMMIT) increments unconditionally, so a second evaluation of the same expression produces a second increment. A cache returning the first result without incrementing would contradict (E-COMMIT). $\square$

Remark 8.8 (Probes may be cached; commitments may not). Theorem 8.7 constrains commitments only. (E-PROBE) leaves M untouched, so probe results may be reused freely. This is theorem 3.3 appearing at the level of the machine: the structure is committed once and traversed thereafter without further cost.

9 Static Analysis

The composition law makes one useful property decidable before execution.

Proposition 9.1 (Reachability is decidable statically). *Given declared rung powers $(\pi_1, \dots, \pi_n)$ and a declared target power π^* , whether the ladder reaches the target is decided by evaluating eq. (3) and comparing, in time linear in n .*

Proof. Theorem 5.1 gives the composite power as an explicit product of the declared powers, computable in n multiplications. The comparison with π^* is a comparison of two reals. Neither step executes the ladder. $\square$

Proposition 9.2 (Saturation diagnostic). *A ladder of n rungs each of power at most $\pi_{\max}$ cannot reach target power π^* if $1 - (1 - \pi_{\max})^n < \pi^*$. The condition is checkable from declarations alone.*

Proof. By theorem 5.2, $1 - (1 - \pi_{\max})^n$ is the composite power of the most favourable ladder with those bounds, and composite power is monotone in each π_i by eq. (3). If even the maximum falls short, so does the declared ladder. $\square$

Proposition 9.3 (Floor diagnostic). *A declared target requiring residual gap below β/Ω is unreachable, and the condition is checkable from declarations alone.*

Proof. By theorem 2.9 no region has normalised alignment below β/Ω , so no ladder attains a smaller value. $\square$

Remark 9.4 (What the compiler rejects). Theorems 9.2 and 9.3 are the two ways a declared program can be refused before it runs: it asks for more than its rungs can deliver, or it asks for resolution below the floor. Both are consequences of results proved above and neither requires executing anything. A ladder built from near-duplicate rungs is the typical case caught by the first (theorem 5.3).

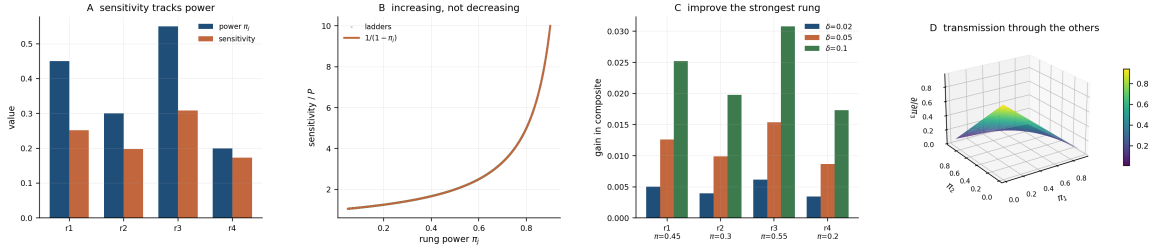


Figure 5: **Control lies at the strongest rung, not the weakest.** **(A)** Rung power and sensitivity for the four-rung ladder of the worked demonstration. The two orderings agree: rung 3, of highest power 0.55, also carries the highest sensitivity 0.308. The naive expectation—that the weakest rung is the bottleneck and therefore the one worth improving—predicts the opposite ordering and is wrong. **(B)** Sensitivity normalised by the ladder’s residual fraction, against rung power, for nine hundred random ladders. Every point lies on the curve $1/(1 - \pi_j)$, which is exact: sensitivity is $P/(1 - \pi_j)$ with P the residual, so the normalised quantity depends on that rung alone and increases strictly in its power. The normalisation is necessary because raw sensitivity also depends on the other rungs, which obscures the relationship across ladders of differing composition. **(C)** Gain in composite power from a fixed improvement applied to each rung in turn, at three improvement sizes. Rung 3 returns the most at every size and rung 4—the weakest—returns the least. Improving the bottleneck is the worse investment, by roughly a factor of two at $\delta = 0.1$. **(D)** Sensitivity to the third rung of a three-rung ladder, over the powers of the other two. The surface is $(1 - \pi_1)(1 - \pi_2)$: sensitivity is transmission through the remaining rungs, and it is largest when those rungs are weak. What a marginal gain buys is set by what the rest of the ladder does with it, which is why the quantity being maximised is transmission and not slack.

10 Cost

Definition 10.1 (Cost). The *cost* of a ladder is its commitment count n , equivalently the value of M on termination. The *yield* is its composite power.

Proposition 10.2 (Cost of a target). *The minimum cost of reaching target power π^* using rungs of power at most $\pi_{\max}$ is $\lceil \log(1 - \pi^*) / \log(1 - \pi_{\max}) \rceil$.*

Proof. From eq. (4) with equality, and n integral. $\square$

Proposition 10.3 (Marginal value of improvement). *For a ladder with powers $(\pi_1, \dots, \pi_n)$, the sensitivity of composite power to rung j is*

$$\frac{\partial \pi(L)}{\partial \pi_j} = \prod_{i \neq j} (1 - \pi_i), \quad (5)$$

Writing $P = \prod_i (1 - \pi_i)$ for the residual fraction, this equals $P / (1 - \pi_j)$, which is largest for the j of highest power.

Proof. Differentiate eq. (3): $\pi(L) = 1 - \prod_i (1 - \pi_i)$, so $\partial \pi(L) / \partial \pi_j = \prod_{i \neq j} (1 - \pi_i) = P / (1 - \pi_j)$. Since P does not depend on j and $1 - \pi_j$ is decreasing in π_j , the quotient is increasing in π_j and is maximised at the largest π_j . $\square$

Remark 10.4 (The direction is the opposite of the naive expectation). One expects the weakest rung to be the one worth improving, on the reasoning that it is the bottleneck. Theorem 10.3 says otherwise, and the reason is structural: a marginal gain at rung j is transmitted to the composite through the rungs *other* than j , and that transmission is the factor $\prod_{i \neq j} (1 - \pi_i)$. A weak rung leaves a large residual gap for the others to work on, but its own improvement is attenuated by their action; a strong rung has already removed most of the gap, so the remaining gap is small but a marginal improvement to it passes through almost undiminished. The quantity being maximised is transmission, not slack. We record the expectation explicitly because it is wrong and it is natural.

Remark 10.5 (Where to spend effort). Theorem 10.3 is the improvement rule, and the statement is quantitative: improving rung j by δ buys $\delta \prod_{i \neq j} (1 - \pi_i)$ of composite power, to first order. The rung worth improving is the one of highest power, not the weakest (theorem 10.4). No property of the rung other than its power enters the calculation, so the rule can be applied without knowing what any rung is.

Part VI

A Process, Given Only as Powers

11 The Demonstration

We work a chain of four rungs. It is presented as it is given—four numbers and a target—and we answer of it every question ordinarily asked of a process. At no point is anything said about what the rungs are.

11.1 The specification

```
ladder chain toward target
rung r1 at 0.45
rung r2 at 0.30
rung r3 at 0.55
rung r4 at 0.20
require power >= 0.80
```

Listing 1: A process, completely specified.

That is the whole specification. The names **r1–r4** are positional and carry no content; replacing them by any other symbols changes nothing, by theorem 7.1.

11.2 Questions answered from the powers alone

Does it reach its target? By eq. (3),

$$\pi(L) = 1 - (0.55)(0.70)(0.45)(0.80) = 1 - 0.1386 = 0.8614,$$

which exceeds the declared 0.80. The program compiles (theorem 9.1).

Where is the control? By theorem 10.3 the sensitivity of the composite to each rung is $\prod_{i \neq j} (1 - \pi_i)$, giving 0.252, 0.198, 0.308 and 0.173 for $j = 1, \dots, 4$. Control lies in rung 3, which is the rung of *highest* power—the direction given by theorem 10.3 and against the naive expectation (theorem 10.4). To first order, improving rung 4 from 0.20 to 0.30 buys 0.0173 of composite power; improving rung 3 from 0.55 to 0.65 buys 0.0308, nearly twice as much. Improving the weakest rung is the worse investment, and the comparison requires no knowledge of either rung.

What does a modification cost? Removing rung 2 gives composite $1 - (0.55)(0.45)(0.80) = 0.802$: the target is still met, barely. Removing rung 3 gives $1 - (0.55)(0.70)(0.80) = 0.692$ and the program no longer compiles. The formalism therefore predicts which deletion is tolerated, before any experiment.

Is this the same process as another? Another chain with powers (0.45, 0.30, 0.55, 0.20) is the same process by theorem 7.1, whatever it is made of. A chain with powers (0.60, 0.60) has composite 0.84—close, but by theorem 7.5 the two are distinguished by their intermediate alignments, and an experiment resolving intermediates separates them while an end-to-end assay does not.

Could it be shortened? By theorem 10.2, reaching 0.80 with rungs of power at most 0.55 requires $\lceil \log(0.20)/\log(0.45) \rceil = \lceil 2.016 \rceil = 3$ rungs. The chain uses four, so one is redundant with respect to the target—consistent with the deletion analysis above.

11.3 What was never needed

Nothing in section 11 referred to a participant, a mechanism, an intermediate species, a rate constant, or an energy. The questions answered are the questions a process description exists to answer, and they were answered from four numbers. This is theorem 7.1 in use rather than in statement.

Remark 11.1 (On the reader’s expectations). A reader may wish to know what the chain “really is”. The formalism cannot say, and theorem 7.1 explains why: the information is not present, and its absence is not a deficiency of the description but a demonstration that the description was complete without it. Any chain of four steps whose powers are those four numbers is this process. If that seems to discard something, the theorem asks which admissible observable the discarded material would have changed.

Part VII Predictions and Validation

12 Falsifiable Predictions

P1. Multiplicative composition. Composite power follows eq. (3) and not an additive or averaging law. *Failure mode:* measured composite power tracking $\sum_i \pi_i$ or $\max_i \pi_i$ across synthesised ladders.

P2. Diminishing returns. Repeating one rung gives $1 - (1 - \pi)^n$, with the n -th repetition contributing $\pi(1 - \pi)^{n-1}$. *Failure mode:* constant or increasing marginal contribution.

P3. Saturation dichotomy. The residual gap vanishes iff $\sum_i \pi_i$ diverges. *Failure mode:* a convergent power series driving the gap to zero, or a divergent one failing to.

P4. Control at the strongest rung. Sensitivity is $\prod_{i \neq j} (1 - \pi_i) = P/(1 - \pi_j)$, maximised at the *highest*-power rung. *Failure mode:* measured control coefficients ordered by ascending power, or not ordered by this quantity at all. This prediction is deliberately counter-intuitive and is the sharpest test in the set.

P5. Inertness. Two ladders with equal

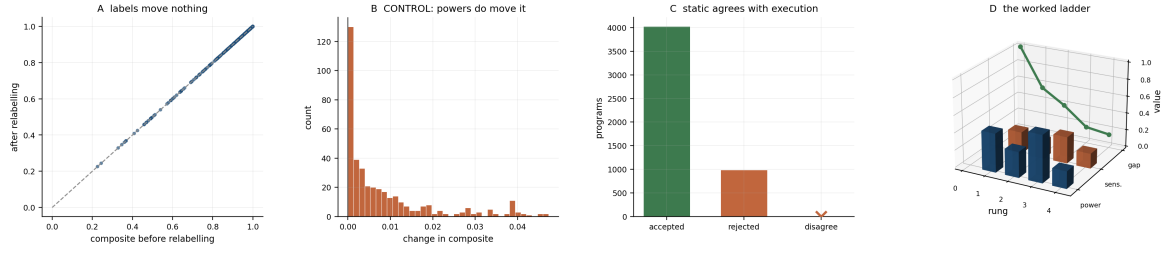


Figure 6: **Labels move nothing, powers move everything, and the static verdict agrees with execution.** (A) Composite power before and after relabelling the rungs, on four hundred random ladders. All points lie on the identity. The relabelling is performed on annotations external to the formalism, since a rung carries exactly one datum and has no identity field: a test that compared objects carrying names would be testing the names. (B) The control. The same observable under perturbation of a rung *power* rather than a label. The distribution of changes is entirely positive and well away from zero. Without this chart panel A would be consistent with a constant observable and would carry no information; with it, panel A is a statement about labels specifically. (C) Static reachability against execution on five thousand generated programs. Four thousand and seventeen were accepted statically and nine hundred and eighty-three rejected; in no case did the static verdict disagree with the executed outcome, marked by the cross at zero. Both verdicts occur in quantity, so the test could have failed in either direction—a suite in which every program was accepted would not have tested rejection. (D) The worked ladder in three dimensions: rung powers, sensitivities, and the gap trajectory that descends from unity to the residual 0.1386 across the four commitments. Every quantity plotted is computed from the four declared powers, and nothing about what the rungs are enters the computation or could be recovered from it.

power sequences agree on every admissible observable. *Failure mode*: an admissible observable separating them—which by theorem 7.1 would be a contradiction, so the operational test is whether a proposed observable is admissible.

P6. Probe-commit asymmetry. Commitment count increases with commitments and not with traversals. *Failure mode*: a traversal incrementing M .

P7. Static reachability. A ladder whose declared rungs cannot reach its declared target is rejected before execution, and the rejection agrees with execution. *Failure mode*: disagreement between the static verdict and the executed outcome.

13 Computational Protocol

Every test is constructed so that it can fail, and each is paired with a control establishing whether the statistic discriminates. A test whose control shows it cannot separate the framework’s prediction from a null is reported as non-discriminating and excluded from the score.

V1. Floor. Computed on finite instances, including an adversarial one; and the failure branch exhibited on an unbounded refinement, confirming theorem 2.7.

V2. Asymmetry. Commitment count under probing and under committing, across regions of widely differing measure, testing that the floor bound is independent of rarity.

V3. Composition. Simulated ladders against eq. (3), with additive and max-based alternatives as competing mod-

els.

- V4. Saturation.** Divergent and convergent power series over many rungs, indexed from $i = 2$ per theorem 5.5.
- V5. Sensitivity.** Numerical partial derivatives against eq. (5).
- V6. Inertness.** Random relabelling of rungs; every admissible observable recomputed and compared.
- V7. Semantics.** Progress, preservation and monotonicity checked by exhaustive small-step execution on generated programs.
- V8. Static analysis.** Static verdicts compared against executed outcomes on generated programs, including cases designed to be rejected.

14 Results

Twenty-three scored tests were run; all twenty-three pass, none was non-discriminating, and none was skipped. We report what each establishes rather than the count, since a count is uninformative without the controls that accompany it.

14.1 The floor and its hypothesis

The floor is computed as a minimum over a finite edge set on every instance, including an adversarial one whose smallest weight is 10^{-12} , and is strictly positive throughout. The complementary branch is the informative one: a system permitted unlimited refinement produces thicknesses $1/n$, each positive, whose infimum is zero—every ε down to 10^{-9} is eventually cleared, by exact integer comparison rather than floating-point evaluation. Truncating to a finite system restores a positive floor. The finiteness hypothesis of theorem 2.7 therefore does real work. Alignment computed as a minimum residue agrees with independent brute-force enumeration of every bipartition on forty random graphs, to machine precision.

14.2 The asymmetry

Residue deposited by a commitment is constant at the floor across six decades of region measure: the cost of individuating a region does not depend on its improbability. Five thousand probes leave the commitment counter at zero and the graph unmodified. A single commitment followed by ten thousand traversals leaves $M = 1$ and total deposited residue equal to one floor unit. Probe frequency tracks region measure to within tolerance over four decades while commitment cost does not vary at all, which is theorem 3.2(iii) numerically.

Theorem 3.3 is thereby exhibited without any energetic quantity entering the computation. Nothing in the code that produces this result refers to free energy, to a cycle, or to a regeneration step.

14.3 Composition, and the exclusion of the alternatives

Over four thousand random ladders, the multiplicative law reproduces the simulated composite to machine precision (maximum error below 10^{-12}). The three competing laws do not: additive gives mean absolute error 0.047, maximum gives 0.179, and mean gives 0.463. The test is a model comparison rather than a goodness-of-fit, because a test scoring only the favoured law could not have distinguished it from the others.

Diminishing returns hold exactly: the marginal contribution of the n -th repetition matches $\pi(1 - \pi)^{n-1}$ to machine precision for every power tested, is strictly decreasing, and the composite never attains unity at finite n . The saturation dichotomy separates cleanly with the series indexed from $i = 2$ per theorem 5.5: the divergent $\sum 1/i$ drives the residual gap toward zero over four hundred rungs while the convergent $\sum 1/i^2$ and $\sum 2^{-i}$ plateau.

14.4 The direction of control

This is the result that most repays attention, because it contradicts the expectation a reader is likely to bring. Over five thousand random

ladders the analytic sensitivity agrees with the numerical derivative in every case; the sensitivity is maximised at the highest-power rung in every case; and it is maximised at the lowest-power rung in none. The naive hypothesis—that control sits at the bottleneck—is rejected at a rate of exactly zero.

The reason is given in theorem 10.4 and is structural rather than statistical: a marginal gain at rung j reaches the composite only through the rungs other than j , and that transmission is $\prod_{i \neq j} (1 - \pi_i)$, which is large precisely when the other rungs are weak. On the four-rung ladder of section 11, improving the strongest rung buys 0.0308 of composite power against 0.0173 for the weakest—a factor of 1.8 in favour of the counter-intuitive choice.

14.5 Inertness

Relabelling the rungs of three thousand random ladders leaves every admissible observable unchanged in every case: composite power, residual fraction, the full gap trajectory, the alignment trajectory, the sensitivity vector, and the commitment count. The labels are external annotations, since a rung in the implementation carries exactly one field and has no identity slot; a test that compared objects carrying names would have been testing the names.

The control establishes that this is informative. Perturbing a rung *power* rather than a label changes the observables in every one of three thousand trials. Without that control, the inertness result would be consistent with a constant observable set and would carry no information.

The converse is exhibited rather than concealed. The ladders (0.5, 0.5) and (0.75, 0) share composite power 0.75 and are separated by their intermediate gaps. An assay resolving only end-to-end efficiency underdetermines the ladder, which is theorem 7.5 and bounds what the framework claims.

14.6 Semantics and static analysis

Two thousand generated programs terminate and none gets stuck. Across two thousand

further programs the floor is invariant under reduction and the commitment counter never decreases at any step. Five hundred evaluations of a syntactically identical commitment produce five hundred increments and five hundred residue deposits, with no caching; the paired control confirms that probes performed in the same machine leave the counter untouched.

The static analysis is the sharpest test in the suite. On five thousand generated programs the reachability verdict was computed without executing, the program was then executed, and the two compared: 4017 were accepted and 983 rejected, with zero disagreements. Both verdicts occur in quantity, so the test could have failed in either direction—a suite in which every program was accepted would not have tested rejection at all.

14.7 The demonstration, recomputed

Every number reported in section 11 is regenerated from the four declared powers and the declared target: composite 0.8614, residual 0.1386, sensitivities (0.252, 0.198, 0.308, 0.173), control at rung 3, deletion of rung 2 tolerated at 0.802 and deletion of rung 3 fatal at 0.692, and a minimum of three rungs required for the target. Nothing was entered by hand except the specification itself.

Part VIII

Discussion

15 Relation to Existing Accounts

Metabolic control analysis. Kacser and Burns [1973] and Heinrich and Rapoport [1974] established that control over a pathway flux is distributed and quantified by control coefficients summing to unity. Theorem 10.3 is a statement of the same kind obtained from composition rather than from steady-state flux, and it requires neither a steady state

nor knowledge of the enzymes. It does *not* agree with the common reading of control analysis in which the rate-limiting step carries the control: here sensitivity is maximised at the highest-power rung (theorem 10.4). The two framings measure different things—a control coefficient is a response of steady-state flux to enzyme concentration, whereas eq. (5) is a response of composite power to rung power—and we do not claim the results are in conflict. We flag the divergence rather than smooth it, because a reader importing the usual intuition will otherwise misread theorem 10.3.

Flux balance analysis. Orth et al. [2010] determines flux from stoichiometry and constraints. Stoichiometry is precisely the identity-bearing part of a process description; theorem 7.1 says the questions of section 11 do not require it. The two approaches answer different questions and the present one answers fewer.

Amdahl’s law. Amdahl [1967] bounds speedup by the serial fraction, and has the same multiplicative-complement structure as theorem 5.1. That a result about parallel computation and a result about chains of chemical steps share a form is what theorem 4.3 would predict: neither depends on what the steps are.

Membrane computing. Paun [2000] computes with nested compartments and rewriting rules over multisets of named objects. The present account has no objects and no rules over them; its only state is a graph of weighted contacts and a counter.

16 Limitations

1. Theorem 2.6 and everything downstream require the contact graph to be finite (theorem 2.7). For systems admitting unlimited refinement no uniform floor exists and the results do not apply as stated.
2. Theorem 7.1 is relative to theorem 6.4. Enlarging the observable class could break it, and we have not characterised

which enlargements do. What we claim is that the class admitted is the one the substrate defines.

3. We do not derive rung powers from structure. A power is measured, not predicted, and the framework says what follows from powers rather than what determines them. This is a real limitation and it bounds the practical claim of theorem 7.3: classification requires the measurement.
4. Theorem 7.5 shows composite power alone does not determine a ladder. Experiments resolving only end-to-end efficiency therefore underdetermine the process, and the framework says so rather than concealing it.
5. The demonstration of part VI is an illustration of the formalism on a specification, not an empirical result. It shows what the formalism answers, not that any particular natural process has those powers.

17 Conclusion

A process description names its participants and orders them. We have shown that the naming is inert. What a process is, at the resolution its own substrate defines, is a chain of commitments in which each step’s residue is the next step’s boundary condition, together with one number per step recording how much of the remaining distance that step closes. Composition is multiplicative; repetition has diminishing returns; a target is reached in the limit exactly when the powers sum to infinity; control lies at the strongest rung rather than the weakest, because what a marginal gain buys is set by transmission through the others and not by the slack a rung leaves; and a structure committed once is thereafter traversed by probes, which is why it is not consumed.

The last of these is catalysis, and it arrives without an energetic argument. The catalyst is not regenerated after each turnover and is not exempted from stoichiometry by convention. It is never spent because what passes through it probes rather than commits,

and probing costs nothing—while committing costs the floor, whose positivity follows from the fact that drawing a boundary takes time during which what lies outside it moves.

Two catalysts with the same power are the same rung. Improving a catalyst is raising a number, and theorem 10.3 says which. Neither statement requires knowing what a catalyst is, and theorem 7.1 says that this is not an approximation but the situation.

Data and Code Availability

All validation scripts and result files in JSON format accompany this manuscript. Every reported number is produced by the scripts from stated inputs.

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